Tyler Waltner

+1(503)-739-4107 | twaltner@u.rochester.edu | github.com/twaltner251

Education:

University of Rochester & Eastman School of Music | *Rochester, NY*

Bachelor of Science in Computer Science & Bachelor in Jazz Performance *Expected May 2028*

Experience:

Coding Camp Instructor | Coding with Kids – Portland, OR Aug. *2024 – Present*

- Taught engaging coding classes to K–12 students using structured curriculum. Topics include Python, AI/ML, data analysis, and game development.
- Led hands-on projects and supported student problem-solving while maintaining a positive classroom environment.

Jazz Musician & Band Leader | Self-employed – Portland, OR / Rochester, NY May 2020 – Present

- Direct 3 active jazz ensembles (Overbridge, Tyler Waltner Quartet/Trio, Nu Zeta), producing 100+ annual live performances.
- Manage full operational lifecycle for all events, including venue booking, contract negotiation, financial tracking, and musical arrangement.

Technical Projects:

Dynamic CLI Minibatch MLP | *Python, NumPy*

- Engineered a custom Multi-Layer Perceptron from scratch without popular ML frameworks to classify the FashionMNIST dataset.
- Scaled an original 1D-optimized stochastic gradient descent (SGD) repository into a minibatch training pipeline, incorporating He weight initialization, state caching, and dynamic CLI-driven hyperparameter configuration.

NASA NeoW API Visualizer | *React, Three.js, WebGL, HTML/CSS*

- Built a 3D web application fetching and rendering real-time near-Earth asteroid data from NASA's NeoWs API.
- Bridged declarative React with imperative WebGL using useRef, implemented smart caching to prevent rate-limiting, and established a custom garbage-collection pipeline to eliminate GPU memory leaks.

Map Routing & Pathfinding Visualizer | *Java, Java Swing*

- Engineered a geographic routing application utilizing Dijkstra's algorithm and a custom PriorityQueue to compute shortest paths.
- Designed a custom Graph data structure mapping intersections and roads, applying the Haversine formula for accurate real-world coordinate distance calculations.
- Developed a dual-interface system featuring a CLI parser for route execution and a Java Swing/Graphics2D GUI for visual navigation rendering.

Technical Skills:

Coding Languages: Python, Java, C, JavaScript, HTML/CSS, R

Frameworks & Libraries: PyTorch, React, Node.js, JavaFX, NumPy, Matplotlib, Three.js, Pygame

Developer Tools: Git, Docker, VS Code, Visual Studio, Vim

Additional Skills: Bilingual (English/Japanese), Project Management, Musical Instrumentation (Saxophones, Clarinet, Flute, Piano)